

Introduction to Machine Learning

Analytics

- Analysis is focused on understanding the past; what happened and why it happened. Analytics focuses on why it happened and what will happen in the future
- Analytics is the discovery, interpretation, and communication of meaningful patterns in data. It also entails applying data patterns towards effective decision making. In other words, analytics can be understood as the connection between data and effective decision making within an organization. Especially valuable in areas rich with recorded information, analytics relies on the simultaneous application of statistics, computer programming and operations research to quantify performance.

Machine Learning

- Machine learning process is a way to deal with creating man-made reasoning. We use Machine Learning calculation to build up the new calculations and strategies. These enable the machine to gain from the examined information or with experience. Most errands that need insight must have a capacity to initiate new information from encounters. Accordingly, a huge region inside AI is machine learning. This includes the investigation of calculations that can separate data without on-line human direction.
- **Machine learning** is an application of artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. **Machine learning** focuses on the development of computer programs that can access data and use it learn for themselves.

Machine Learning

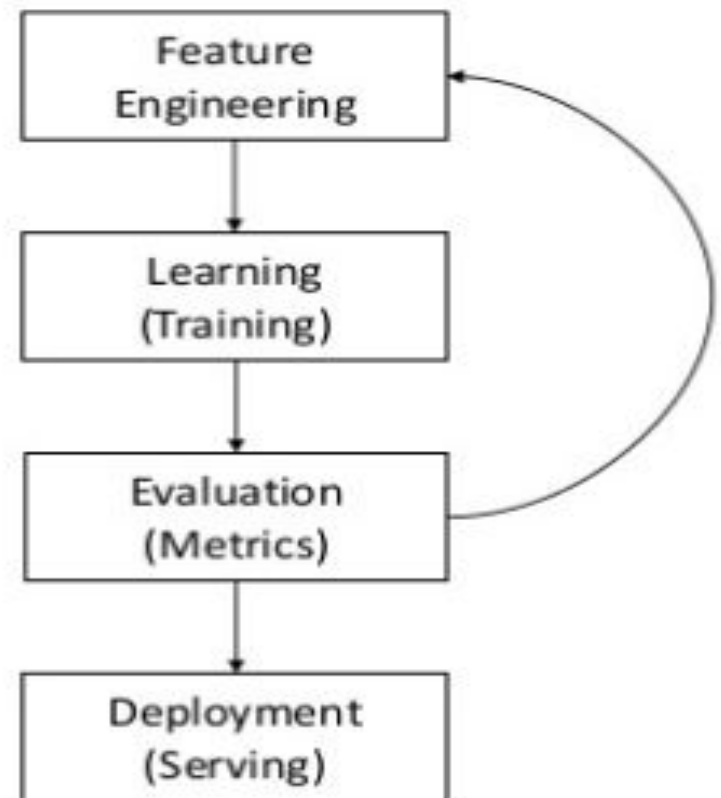
- Machine Learning identifies with the investigation, structure, and improvement of the calculations. These give PCs the ability to learn without being expressly modified. Information Mining begins with unstructured information and attempts to separate learning or intriguing examples. Amid this procedure, we use machine Learning calculations.

ML

- Machine Learning is a field which is raised out of Artificial Intelligence (AI). Applying AI, we needed to assemble better and wise machines. In any case, with the exception of couple of simple errands, for example, finding the most brief way between point A and B, we were not able program increasingly mind boggling and always advancing challenges. There was an acknowledgment that the best way to almost certainly accomplish this assignment was to give machine a chance to gain from itself. This sounds like a youngster gaining from its self. So machine learning was produced as another capacity for PCs. What's more, presently machine realizing is available in such huge numbers of sections of innovation, that we don't understand it while utilizing it.

ML Process

Ability of a machine to improve its own performance through the use of a software that employs artificial intelligence techniques to mimic the ways by which humans seem to learn such as repetition and experience.



ML Process

- Helps in building machines exhibiting intelligent behavior.
- Apart from artificial intelligence it is also used in administration, commerce and industry.
- The most widely known demonstration of this migration is '**DATA MINING**'.

BENEFITS:

- Makes human-computer interaction easier
- Relatively simple to integrate
- Will distinguish your products from others
- Increase customer satisfaction
- Will improve simple-intelligent systems

Applications of Machine learning

APPLICATIONS:

- Medical diagnosis
- Data mining
- Bioinformatics
- Speech and handwriting recognition
- Product categorization
- Inertial measurement unit (IMU)
- Information retrieval



- **Machine learning**, a branch of artificial intelligence, is a scientific discipline concerned with the design and development of algorithms.
- **Computer system** (expert system) which is imbued with decision making ability like a human expert.
- Two parts: a. Knowledge base (database)
b. Inference engine (predicted with certain probability)
- It is a learner (like a small baby) which looks at the examples (obstacles), analyses it using stored data which also includes previous experiences, finds its algorithm and predicts the possible solution with highest probability.
- It keeps updating itself with every obstacle solved, enhancing its performance every time.
- Its algorithms includes different combinations of logic.

Categories of ML

Machine Learning Algorithms *(sample)*

	<u>Unsupervised</u>	<u>Supervised</u>
<u>Continuous</u>	• Clustering & Dimensionality Reduction ○ SVD ○ PCA ○ K-means	• Regression ○ Linear ○ Polynomial • Decision Trees • Random Forests
<u>Categorical</u>	• Association Analysis ○ Apriori ○ FP-Growth • Hidden Markov Model	• Classification ○ KNN ○ Trees ○ Logistic Regression ○ Naive-Bayes ○ SVM

Kinds of Machine Learning

1

Supervised
Learning

2

Unsupervised
Learning

3

Reinforcement
Learning

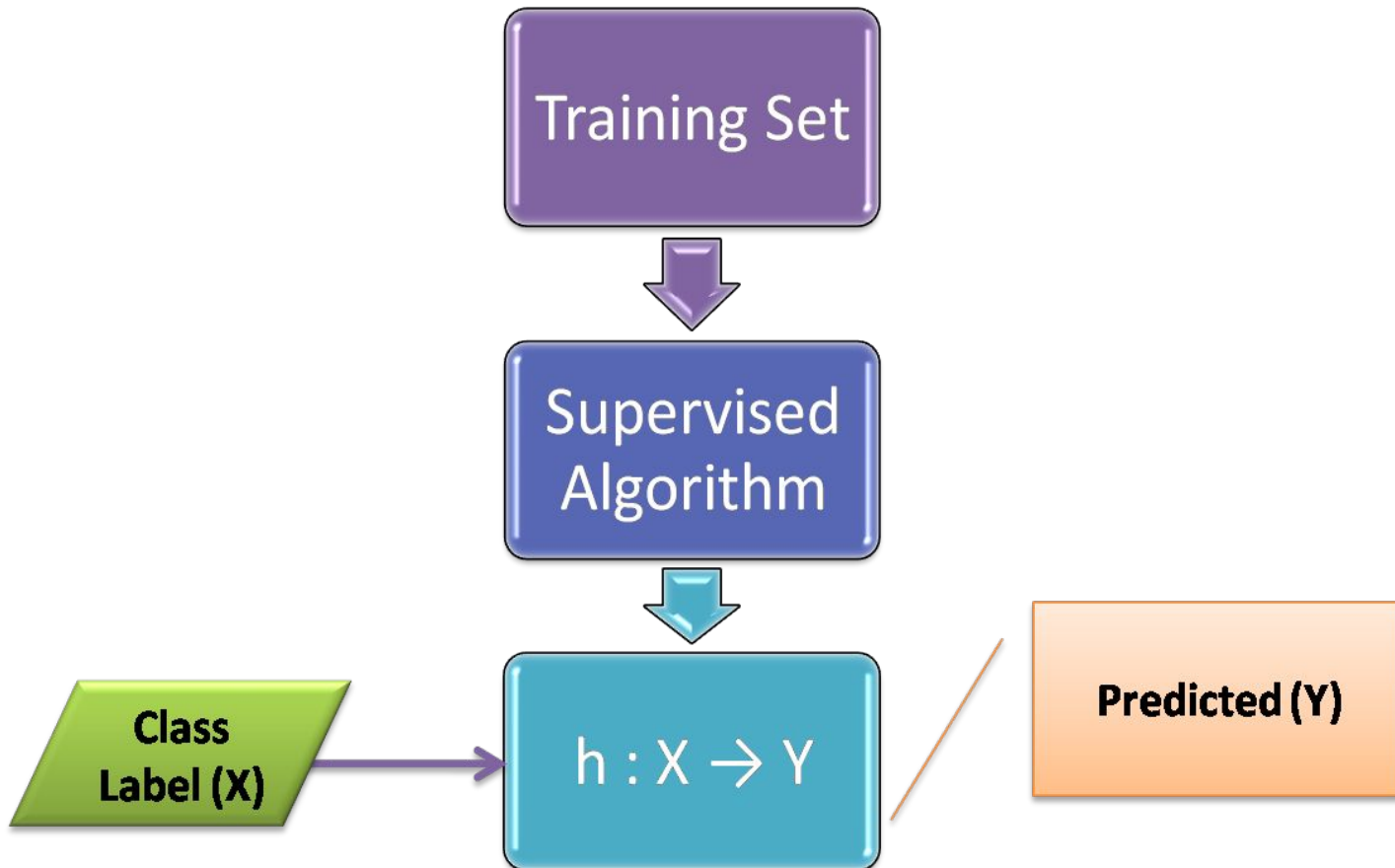
Supervised Learning

- “The outcome or output for the given input is known before itself” and the machine must be able to map or assign the given input to the output. Multiple images of a cat, dog, orange, apple etc here the images are labelled. It is fed into the machine for training and the machine must identify the same. Just like a human child is shown a cat and told so, when it sees a completely different cat among others still identifies it as a cat, the same method is employed here.

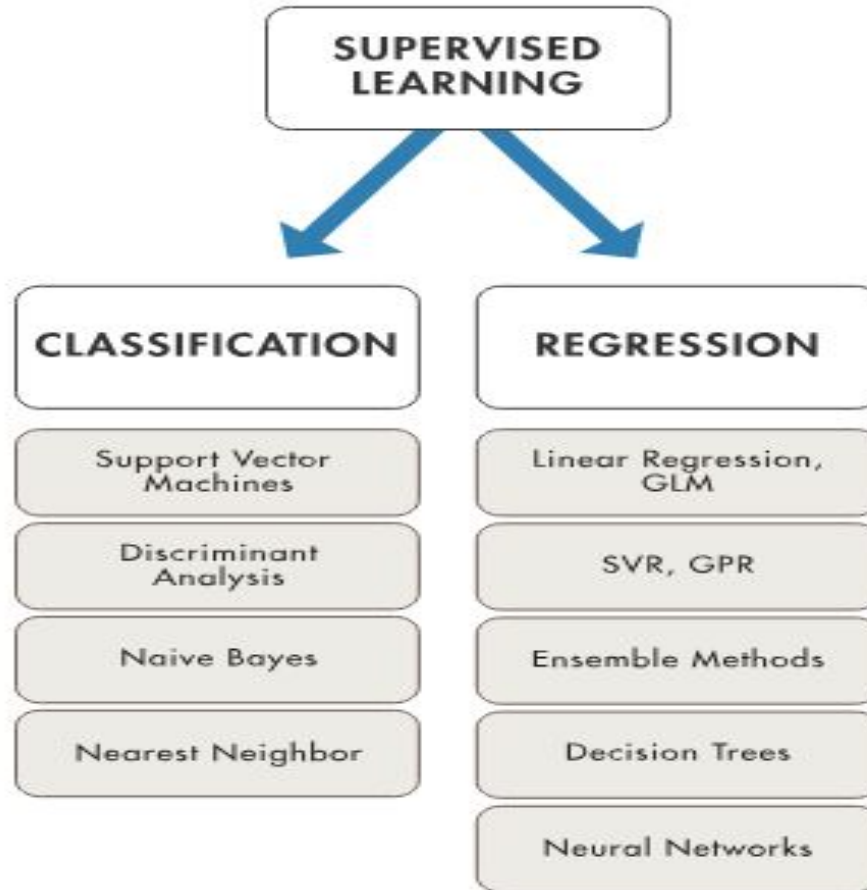
Supervised Learning

- In regulated learning, we are given an informational index and as of now comprehend what our right yield should resemble, having the possibility that there is a connection between the information and the yield. Directed learning issues are additionally sorted into relapse and arrangement issues.
- To portray the directed learning issue somewhat more formally, our objective is, given a preparation set, to get familiar with a capacity $h : X \rightarrow Y$ so that $h(x)$ is a "decent" indicator for the comparing estimation of y . For authentic reasons, this capacity h is known as a theory. Seen pictorially, the procedure is thusly similar to this:

Supervised Learning



Supervised Learning



Unsupervised Learning

- *“The outcome or output for the given inputs is unknown”*, here input data is given and the model is run on it. The image or the input given are grouped together here and insights on the inputs can be found here(which is the most of the real world data available). The main algorithms include ***Clustering algorithms() and*** learning algorithms.

Unsupervised Learning

- Unsupervised learning works by breaking down the information without its marks for the shrouded structures inside it, and through deciding the relationships, and for highlights that really correspond two information things. It is being utilized for bunching, dimensionality decrease, include learning, thickness estimation, and so forth.
- The shrouded structure once in a while called include vector, speaks to the info information such a way, that on the off chance that a similar component vector is being use to recreate the info, at that point one can do that with some satisfactory misfortune.

Unsupervised Learning

- The change in two component vectors of two data sources specifically relative to the difference in the information sources itself. Along these lines this shrouded structure or highlight vector just speaks to highlights in the information that really offer refinement to it.

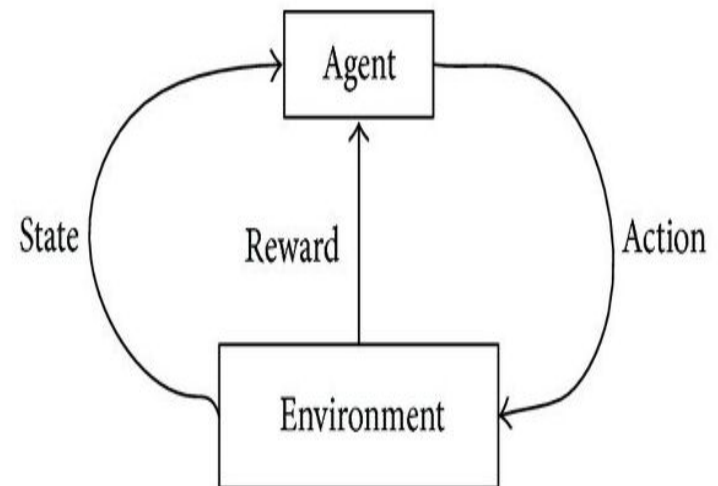
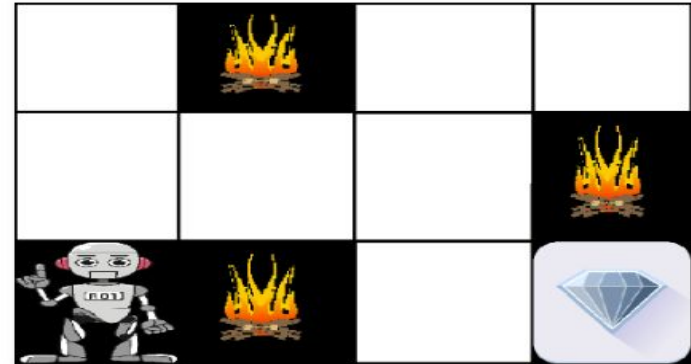


Reinforcement Learning

- Reinforcement learning is a region of Machine Learning. Reinforcement: It is tied in with making reasonable move to expand compensate in a specific circumstance. It is utilized by different programming and machines to locate the most ideal conduct or way it should take in a particular circumstance. Fortification taking in contrasts from the directed learning in a route that in regulated learning the preparation information has the appropriate response key with it so the model is prepared with the right answer itself while in Reinforcement learning, there is no answer yet the fortification specialist chooses what to do to play out the given errand. Without preparing dataset, it will undoubtedly gain from its experience.

Reinforcement Learning

- Precedent : The issue is as per the following: We have a specialist and a reward, with numerous obstacles in the middle. The specialist should locate the most ideal way to achieve the reward. The accompanying issue clarifies the issue all the more effectively.



Reinforcement Learning

- The above picture indicates robot, precious stone and flame. The objective of the robot is to get the reward that is the jewel and dodge the obstacles that is fire. The robot learns by attempting all the conceivable ways and after that picking the way which gives him the reward with the least obstacles. Each correct advance will give the robot a reward and each wrong advance will subtract the reward of the robot. The absolute reward will be determined when it achieves the last reward that is the precious stone.

Reinforcement Learning

- The above picture indicates robot, precious stone and flame. The objective of the robot is to get the reward that is the jewel and dodge the obstacles that is fire. The robot learns by attempting all the conceivable ways and after that picking the way which gives him the reward with the least obstacles. Each correct advance will give the robot a reward and each wrong advance will subtract the reward of the robot. The absolute reward will be determined when it achieves the last reward that is the precious stone.

Supervised Learning Algorithms with applications in Predictive Analytics

- Predictive analytics is driven by predictive modelling. It's more of an approach than a process. Predictive analytics and [machine learning](#) go hand-in-hand, as predictive models typically include a [machine learning](#) algorithm. These models can be trained over time to respond to new data or values, delivering the results the business needs. Predictive modelling largely overlaps with the field of [machine learning](#).
- There are two types of predictive models. They are Classification models, that predict class membership, and Regression models that predict a number. These models are then made up of algorithms. The algorithms perform the data mining and statistical analysis, determining trends and patterns in data. Predictive analytics software solutions will have built in algorithms that can be used to make predictive models. The algorithms are defined as 'classifiers', identifying which set of categories data belongs to.

The most widely used predictive models are:

- **Decision trees:**
Decision trees are a simple, but powerful form of multiple variable analysis. They are produced by algorithms that identify various ways of splitting data into branch-like segments. Decision trees partition data into subsets based on categories of input variables, helping you to understand someone's path of decisions.
- **Regression (linear and logistic)**
Regression is one of the most popular methods in statistics. Regression analysis estimates relationships among variables, finding key patterns in large and diverse data sets and how they relate to each other.
- **Neural networks**
Patterned after the operation of neurons in the human brain, neural networks (also called artificial neural networks) are a variety of deep learning technologies. They're typically used to solve complex pattern recognition problems – and are incredibly useful for analysing large data sets. They are great at handling nonlinear relationships in data – and work well when certain variables are unknown

Other classifiers:

- **Time Series Algorithms:** Time series algorithms sequentially plot data and are useful for forecasting continuous values over time.
- **Clustering Algorithms:** Clustering algorithms organise data into groups whose members are similar.
- **Outlier Detection Algorithms:** Outlier detection algorithms focus on anomaly detection, identifying items, events or observations that do not conform to an expected pattern or standard within a data set.
- **Ensemble Models:** Ensemble models use multiple machine learning algorithms to obtain better predictive performance than what could be obtained from one algorithm alone.
- **Factor Analysis:** Factor analysis is a method used to describe variability and aims to find independent latent variables.
- **Naïve Bayes:** The Naïve Bayes classifier allows us to predict a class/category based on a given set of features, using probability.
- **Support vector machines:** Support vector machines are supervised [machine learning](#) techniques that use associated learning algorithms to analyse data and recognise patterns.
- Each classifier approaches data in a different way, therefore for organisations to get the results they need, they need to choose the right classifiers and models.

Applications of predictive analytics

- For organisations overflowing with data but struggling to turn it into useful insights, predictive analytics and [machine learning](#) can provide the solution. No matter how much data an organisation has, if it can't use that data to enhance internal and external processes and meet objectives, the data becomes a useless resource.
- Predictive analytics is most commonly used for security, marketing, operations, risk and fraud detection. Here are just a few examples of how predictive analytics and [machine learning](#) are utilised in different industries:

Applications of predictive analytics

- **Banking and Financial Services**

In the banking and financial services industry, predictive analytics and [machine learning](#) are used in conjunction to detect and reduce fraud, measure market risk, identify opportunities and much, much more.

- **Security**

With cybersecurity at the top of every business' agenda in 2017, it should come as no surprise that predictive analytics and [machine learning](#) play a key part in security. Security institutions typically use predictive analytics to improve services and performance, but also to detect anomalies, fraud, understand consumer behaviour and enhance data security.

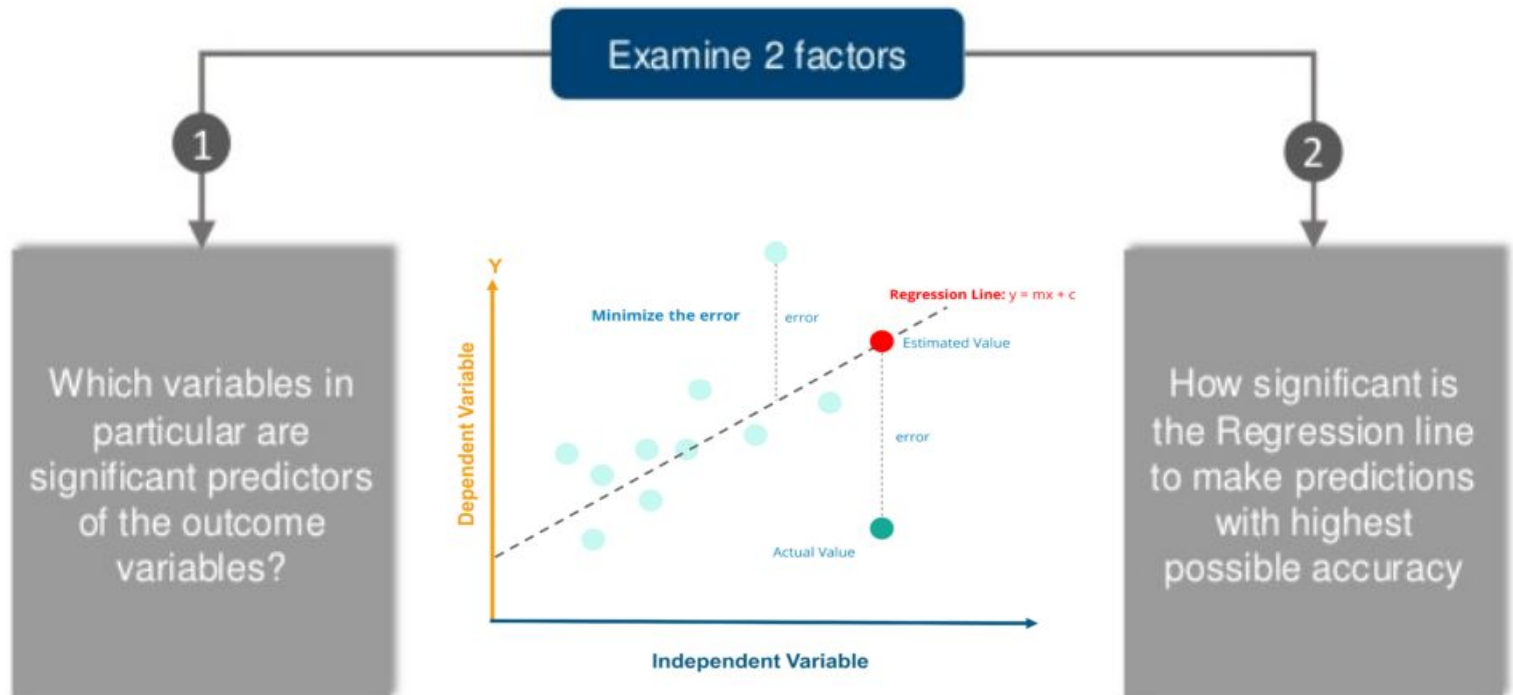
- **Retail**

Retailers are using predictive analytics and [machine learning](#) to better understand consumer behaviour; who buys what and where? These questions can be readily answered with the right predictive models and data sets, helping retailers to plan ahead and stock items based on seasonality and consumer trends – improving ROI significantly.

building a Simple Regression Model

Understanding Linear Regression

Linear Regression is a statistical model used to predict the relationship between independent and dependent variables.



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Regression Equation

The simplest form of a simple linear regression equation with one dependent and one independent variable is represented by:

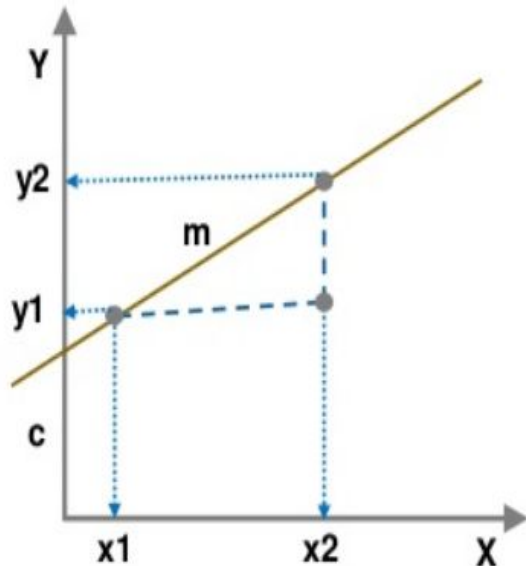
$$y = m * x + c$$

Regression Line, $y = mx + c$ where,

y = Dependent Variable

x = Independent Variable ; c = y-Intercept

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$



y ---> Dependent Variable

x ---> Independent Variable

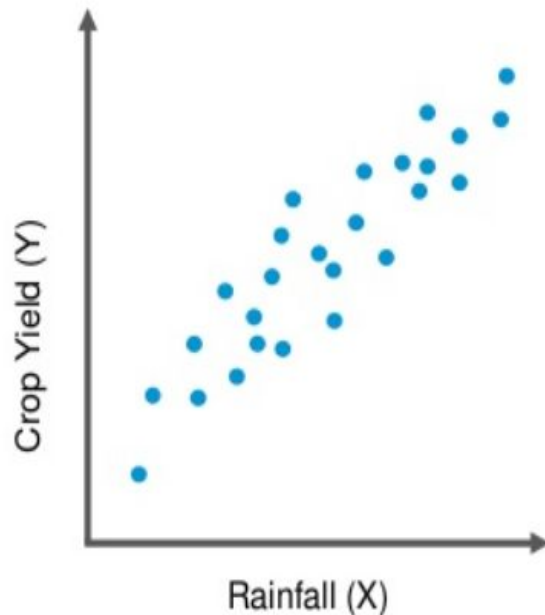
m ---> Slope of the line

c ---> Coefficient of the line

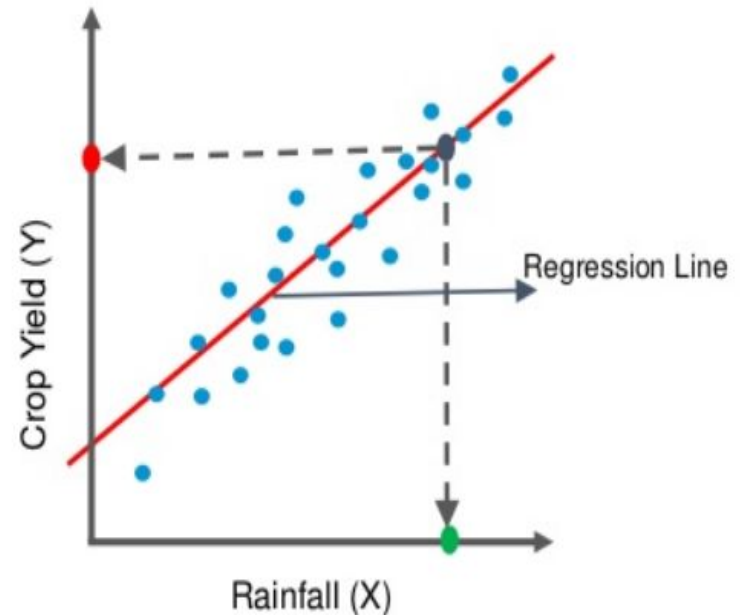
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

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Prediction using the Regression line



Plotting the amount of Crop Yield based on the amount of Rainfall



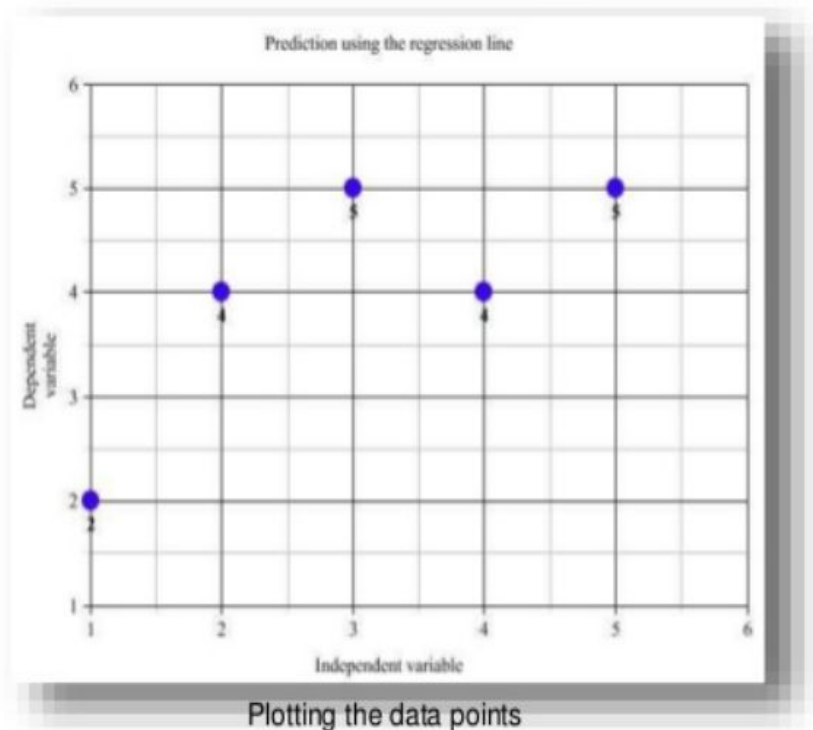
The Red point on the Y axis is the amount of Crop Yield you can expect for some amount of Rainfall (X) represented by Green dot

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Intuition behind the Regression line

Lets consider a sample dataset with 5 rows and find out how to draw the regression line

Independent variable X	Dependent variable Y
1	2
2	4
3	5
4	4
5	5



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Intuition behind the Regression line

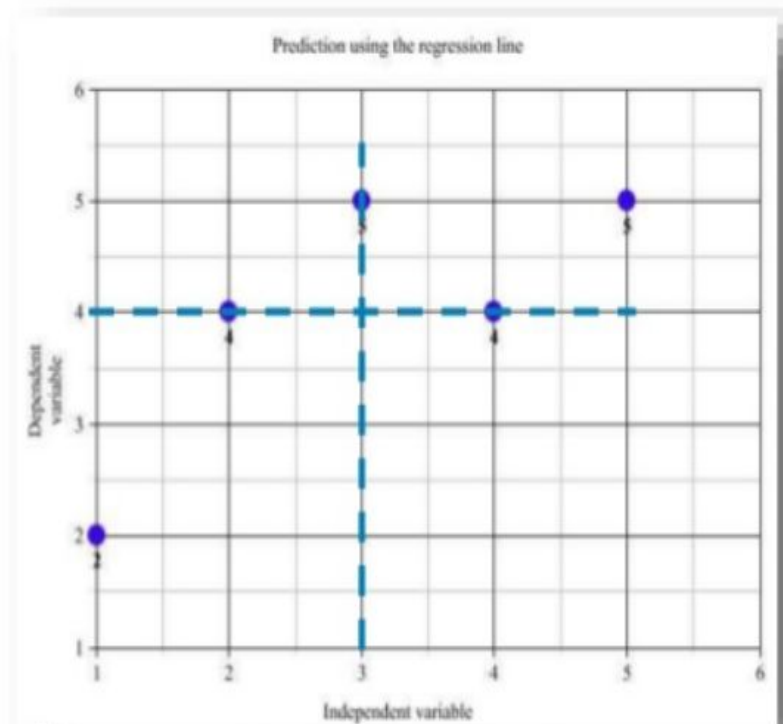
Calculate the mean of X and Y and plot the values

Independent variable	Dependent variable
X	Y
1	2
2	4
3	5
4	4
5	5

Mean

3

4



Plotting the mean of X and Y

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Intuition behind the Regression line

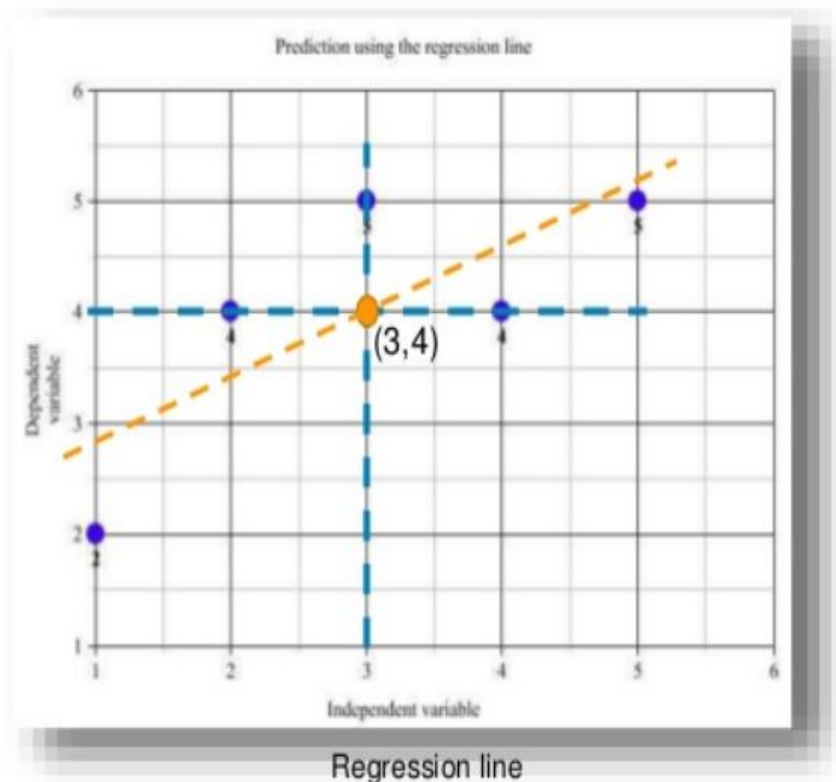
Regression line should ideally pass through the mean of X and Y

Independent variable	Dependent variable
X	Y
1	2
2	4
3	5
4	4
5	5

Mean

3

4



building a Simple Regression Model

Intuition behind the Regression line

Drawing the equation of the Regression line

X	Y	(X ²)	(Y ²)	(X*Y)
1	2	1	4	2
2	4	4	16	8
3	5	9	25	15
4	4	16	16	16
5	5	25	25	25
$\Sigma = 15$	$\Sigma = 20$	$\Sigma = 55$	$\Sigma = 86$	$\Sigma = 66$

$$\begin{aligned}
 Y &= m * X + c \\
 &= 0.6 * 8 + 2.2 \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 a &= \frac{\Sigma (Y)(\Sigma X^2) - (\Sigma X)(\Sigma XY)}{n(\Sigma X^2) - (\Sigma X)^2} \\
 b &= \frac{n(\Sigma XY) - (\Sigma X)(\Sigma Y)}{n(\Sigma X^2) - (\Sigma X)^2}
 \end{aligned}$$

Linear equation is represented as $Y = m X + c$

$$m = \frac{((n * \Sigma X * Y) - (\Sigma X) * (\Sigma Y))}{((n * \Sigma X^2) - (\Sigma X)^2)} = \frac{((5 * 66) - (15 * 20))}{((5 * 55) - (225))} = 0.6$$

$$c = \frac{((\Sigma Y) * (\Sigma X)) - ((\Sigma X)^2 * (\Sigma Y) / n)}{((n * \Sigma X^2) - (\Sigma X)^2)} = 2.2$$

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$$Y = a + b \cdot x$$

Drawing the equation of the Regression line

X	Y	(X ²)	(Y ²)	(X+Y)
1	2	1	4	2
2	4	4	16	8
3	5	9	25	15
4	4	16	16	16
5	5	25	25	25
$\Sigma = 15$	$\Sigma = 20$	$\Sigma = 55$	$\Sigma = 86$	$\Sigma = 66$

$$a = \frac{(\Sigma Y)(\Sigma X^2) - (\Sigma X)(\Sigma XY)}{n(\Sigma X^2) - (\Sigma X)^2}$$

$$b = \frac{n(\Sigma XY) - (\Sigma X)(\Sigma Y)}{n(\Sigma X^2) - (\Sigma X)^2}$$

$$Y = m \cdot x + c$$

$$m = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2} = \frac{(5)(66) - (15)(20)}{(5)(55) - (15)^2} = \frac{330 - 300}{275 - 225} = \frac{30}{50} = 0.6$$

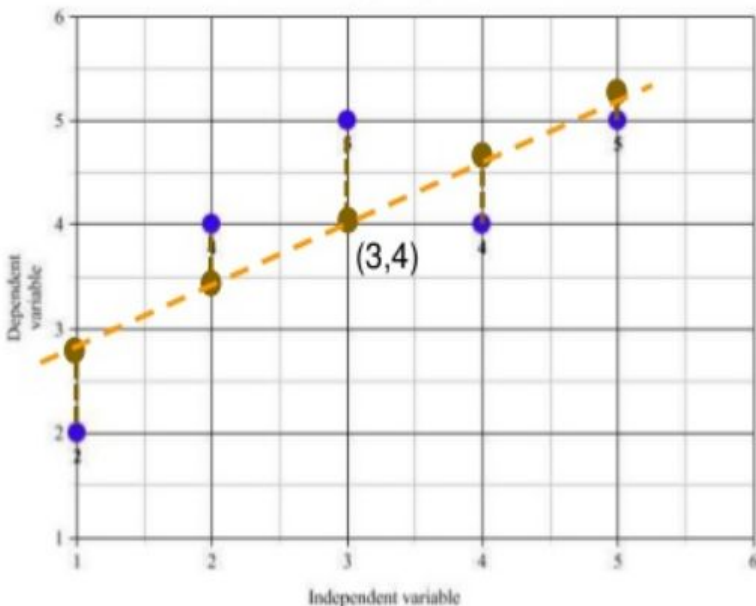
$$c = \frac{(\sum Y)(\sum X^2) - (\sum X)(\sum XY)}{n(\sum X^2) - (\sum X)^2} = \frac{(20)(55) - (15)(66)}{5(55) - (15)^2} = \frac{1100 - 990}{275 - 225} = \frac{110}{50} = 2.2$$

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Intuition behind the Regression line

Lets find out the predicted values of Y for corresponding values of X using the linear equation where $m=0.6$ and $c=2.2$

Prediction using the regression line



Y_{pred}

$$Y = 0.6 * 1 + 2.2 = 2.8$$

$$Y = 0.6 * 2 + 2.2 = 3.4$$

$$Y = 0.6 * 3 + 2.2 = 4$$

$$Y = 0.6 * 4 + 2.2 = 4.6$$

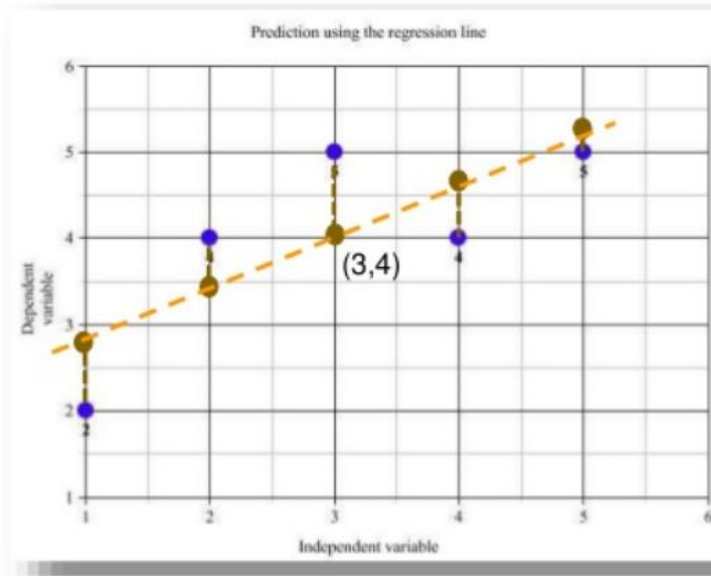
$$Y = 0.6 * 5 + 2.2 = 5.2$$

Here the blue points represent the **actual Y values** and the brown points represent the **predicted Y values**. The distance between the actual and predicted values are known as **residuals or errors**. The best fit line should have the least sum of squares of these errors also known as **e square**.

building a Simple Regression Model

Intuition behind the Regression line

Lets find out the predicted values of Y for corresponding values of X using the linear equation where $m=0.6$ and $c=2.2$



X	Y	Y_{pred}	$(Y - Y_{\text{pred}})$	$(Y - Y_{\text{pred}})^2$
1	2	2.8	-0.8	0.64
2	4	3.4	0.6	0.36
3	5	4	1	1
4	4	4.6	-0.6	0.36
5	5	5.2	-0.2	0.04

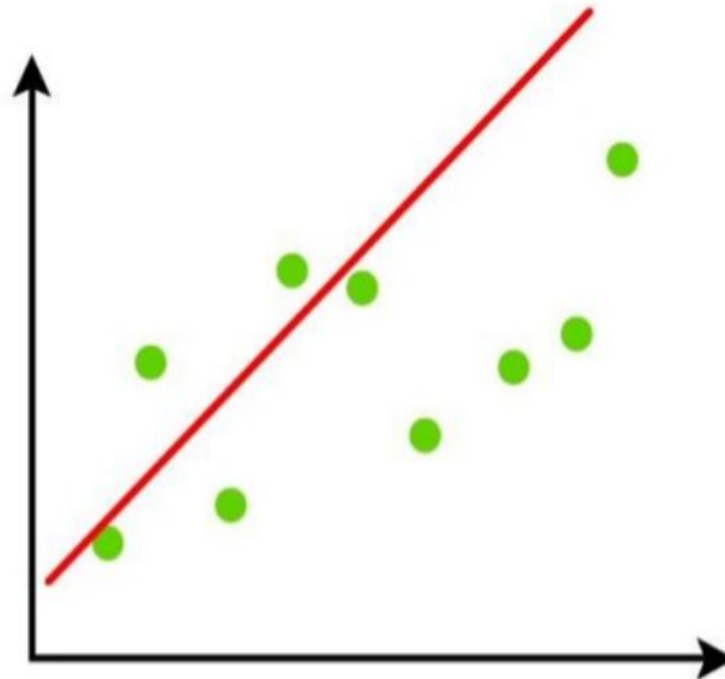
$$\Sigma = 2.4$$

The sum of squared errors for this regression line is 2.4. We check this error for each line and conclude the best fit line having the least square value.

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Finding the Best fit line

Minimizing the Distance: There are lots of ways to minimize the distance between the line and the data points like Sum of Squared errors, Sum of Absolute errors, Root Mean Square error etc.



We keep moving this line through the data points to make sure the Best fit line has the least square distance between the data points and the regression line

building a Simple Regression Model

- Import

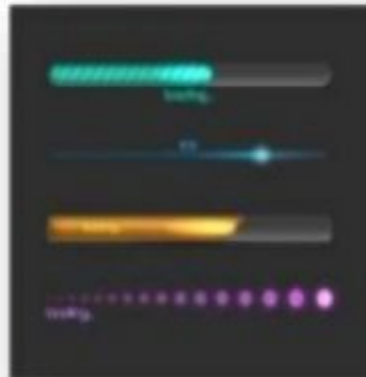
1. Import the libraries:



```
# Importing the Libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
%matplotlib inline
```

building a Simple Regression Model

2. Load the Dataset and extract independent and dependent variables:



```
# Importing the dataset and Extracting the Independent and Dependent variables
companies = pd.read_csv('C:/Users/avijeet.biswal/Desktop/1000_Companies.csv')
X = companies.iloc[:, :-1].values
y = companies.iloc[:, 4].values
```

```
companies.head()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

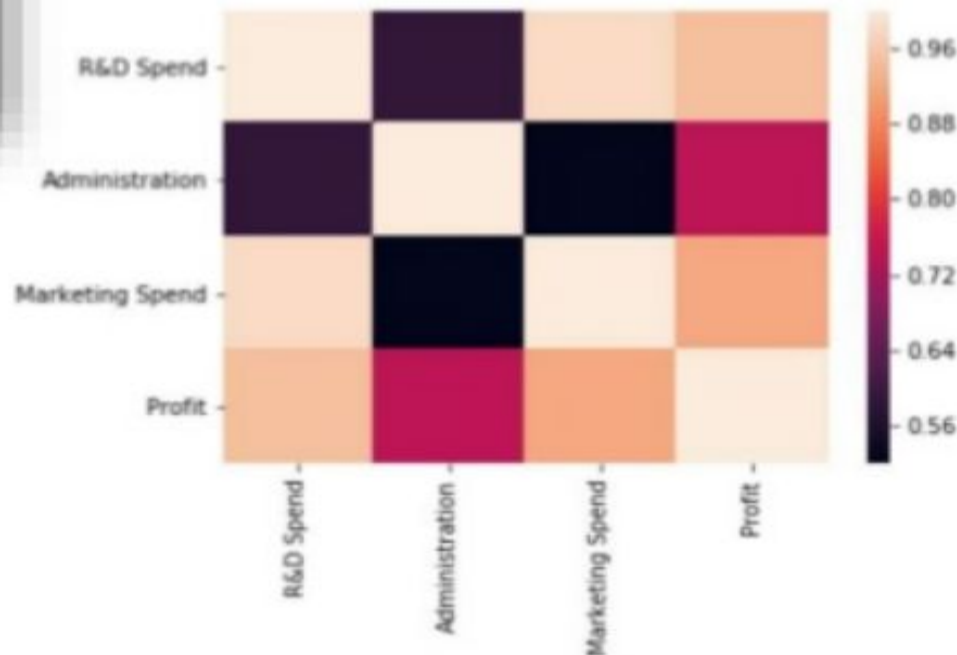
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3. Data Visualization:



```
# Data Visualisation  
# Building the Correlation matrix  
sns.heatmap(companies.corr())
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x8aae1c6400>
```



building a Simple Regression Model

4. Encoding Categorical Data:



```
# Encoding categorical data
from sklearn.preprocessing import LabelEncoder, OneHotEncoder
labelencoder = LabelEncoder()
X[:, 3] = labelencoder.fit_transform(X[:, 3])
onehotencoder = OneHotEncoder(categorical_features = [3])
X = onehotencoder.fit_transform(X).toarray()
```

5. Avoiding Dummy Variable Trap:



```
# Avoiding the Dummy Variable Trap
X = X[:, 1:]
```

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6. *Splitting the data into Train and Test set:*



```
# Splitting the dataset into the Training set and Test set
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 0)
```

7. *Fitting Multiple Linear Regression Model to Training set:*



```
# Fitting Multiple Linear Regression to the Training set
from sklearn.linear_model import LinearRegression
model_fit = LinearRegression()
model_fit.fit(X_train, y_train)

LinearRegression(copy_X=True, fit_intercept=True, n_jobs=1, normalize=False)
```

building a Simple Regression Model

8. Predicting the Test set results:



```
# Predicting the Test set results
y_pred = regressor.predict(X_test)
y_pred

array([[ 89790.61532915,  88427.07187361,  94894.67836972,
        175680.86725611,  83411.73042088, 110571.90200074,
        132145.22936439,  91473.37719686, 164597.05380606,
         53222.82667401,  66950.19050989, 150566.43987005,
        126915.20858596,  59337.8597105 , 177513.91053062,
         75316.28143051, 118248.14406603, 164574.40699902,
        170937.2898107 , 182069.11645084, 118845.03252689,
         85669.95112229, 180992.59396144,  84145.08220145,
        105005.83769214, 101233.56772747,  53831.07669091,
         56881.41475224,  68896.39346905, 210040.00765883,
        120778.72270894, 111724.87157654, 101487.90541518,
        137959.02649624,  63969.95996743, 108857.91214126,
        186014.72531988, 171442.64130747, 174644.26529205,
        117671.49128195,  96731.37857433, 165452.25779409,
        107724.34331255,  50194.54176913, 116513.89532179,
         58632.4898682 , 158416.4682761 ,  78541.48521609,
        159727.66671743, 131137.87699644, 184880.70924516,
        174609.0826688 ,  93745.66352059,  78341.13383418,
        180745.9043908 ,  84461.61490552, 142900.90602903,
        170618.44098397,  84365.09530839, 105307.3716218 ,
        141660.07290787,  52527.34340442, 141842.9626416 ,
        139176.27973195,  98294.52669666, 113586.86790969,
```

building a Simple Regression Model

9. Calculating the Coefficients and Intercepts:



```
# Calculating the Coefficients
print(regressor.coef_)

[ -8.80536598e+02 -6.98169073e+02  5.25845857e-01  8.44390881e-01
  1.07574255e-01]

# Calculating the Intercept
print(regressor.intercept_)

-51035.229724
```

10. Evaluating the model:



```
# Calculating the R squared value
from sklearn.metrics import r2_score
r2_score(y_test, y_pred)

0.91126958922688628
```

R squared value of 0.91 proves the model is a good model

How To Measure The Accuracy Of Predictive Models

- When developing predictive models and algorithms, whether linear regression or [ARIMA models](#) it is important to quantify how well the model fits to the future observations. One of the simplest methods of calculating how correct a model is uses the error between the predicted value and the actual value. From there, there are several methodologies that take this difference and further exploit meaning from it. Quantifying the accuracy of an algorithm is an important step to justifying the usage of the algorithm in product.

MAE

- The Mean absolute error represents the average of the absolute difference between the actual and predicted values in the dataset. It measures the average of the residuals in the dataset.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}|$$

Where,

$\hat{y}$ – predicted value of y

$\bar{y}$ – mean value of y

MSE

- Mean Squared Error represents the average of the squared difference between the original and predicted values in the data set. It measures the variance of the residuals.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

RMSE

- Root Mean Squared Error is the square root of Mean Squared error. It measures the standard deviation of residuals.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2}$$

R Squared

- The coefficient of determination or R-squared represents the proportion of the variance in the dependent variable which is explained by the linear regression model. It is a scale-free score i.e. irrespective of the values being small or large, the value of R square will be less than one.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y})^2}{\sum(y_i - \bar{y})^2}$$

Adjusted R Squared

- Adjusted R squared is a modified version of R square, and it is adjusted for the number of independent variables in the model, and it will always be less than or equal to R^2 . In the formula below n is the number of observations in the data and k is the number of the independent variables in the data.

$$R_{adj}^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - k - 1} \right]$$

Evaluation Measures

Differences among these evaluation metrics

- Mean Squared Error(MSE) and Root Mean Square Error penalizes the large prediction errors vi-a-vis Mean Absolute Error (MAE). However, RMSE is widely used than MSE to evaluate the performance of the regression model with other random models as it has the same units as the dependent variable (Y-axis).
- MSE is a differentiable function that makes it easy to perform mathematical operations in comparison to a non-differentiable function like MAE. Therefore, in many models, RMSE is used as a default metric for calculating Loss Function despite being harder to interpret than MAE.
- MAE is more robust to data with outliers.
- The lower value of MAE, MSE, and RMSE implies higher accuracy of a regression model. However, a higher value of R square is considered desirable.

Evaluation Measures

- R Squared & Adjusted R Squared are used for explaining how well the independent variables in the linear regression model explain the variability in the dependent variable. R Squared value always increases with the addition of the independent variables which might lead to the addition of the redundant variables in our model. However, the adjusted R-squared solves this problem.
- Adjusted R squared takes into account the number of predictor variables, and it is used to determine the number of independent variables in our model. The value of Adjusted R squared decreases if the increase in the R square by the additional variable isn't significant enough.
- For comparing the accuracy among different linear regression models, RMSE is a better choice than R Squared.

Conclusion

Conclusion

Therefore, if comparing the prediction accuracy among different linear regression (LR) models then RMSE is a better option as it is simple to calculate and differentiable. However, if your dataset has outliers then choose MAE over RMSE.

Besides, the number of predictor variables in a linear regression model is determined by adjusted R squared, and choose RMSE over adjusted R squared if you care about evaluating prediction accuracy among different LR models.